

Random Forest – Final Direct Forecasts

Start a new file so dont need to do the cross validation once again lol

0. Packages

```
R_packages <- c(
  "readr", "dplyr", "lubridate",
  "ranger", "tidyr", "tibble", "knitr"
)

options(repos = c(CRAN = "http://cran.rstudio.com"))

if (!requireNamespace("librarian", quietly = TRUE)) {
  install.packages("librarian")
}

librarian::shelf(R_packages)
```

```
## Warning: package 'readr' was built under R version 4.5.2
```

```
set.seed(12345)
```

1. Data

```
df_ml <- read_csv("balanced_can_md.csv", show_col_types = FALSE) %>%
  mutate(
    Date      = as.Date(Date),
    GDP_per_w = GDP_new - EMP_CAN
  ) %>%
  arrange(Date)

targets <- c("GDP_per_w", "CPI_ALL_CAN", "NHOUSE_P_CAN")

features_large <- c(
  "GDP_per_w", "CPI_ALL_CAN", "NHOUSE_P_CAN", "IP_new",
  "BSI_new", "UNEMP_CAN", "EMP_CONS_CAN", "hstart_CAN_new",
  "CRED_HOUS", "CRED_HOUS_MORT", "BANK_RATE_L", "TBILL_3M",
  "Exp_BP_new", "Imp_BP_new", "IPPI_METAL_CAN", "OILP_new",
  "WTISPLC"
)

setdiff(features_large, names(df_ml))
```

```
## character(0)
```

2. HELPER

```
build_lagged_panel <- function(df, predictors, targets, p, h) {
  dat <- df %>%
    dplyr::select(Date, dplyr::all_of(unique(c(predictors, targets)))) %>%
    arrange(Date)

  # add lags for each predictor
  for (lag in 1:p) {
    for (v in predictors) {
      new_name <- paste0("L", lag, "_", v)
      dat[[new_name]] <- dplyr::lag(dat[[v]], lag)
    }
  }

  # add leads for each target at horizon h
  for (v in targets) {
    lead_name <- paste0("F", h, "_", v)
    dat[[lead_name]] <- dplyr::lead(dat[[v]], h)
  }

  lag_cols <- grep("^L[0-9]+_", names(dat), value = TRUE)
  y_cols <- paste0("F", h, "_", targets)

  dat_final <- dat %>%
    dplyr::select(dplyr::all_of(c(lag_cols, y_cols))) %>%
    tidyr::drop_na()

  list(
    X = as.matrix(dat_final[, lag_cols, drop = FALSE]),
    Y = as.matrix(dat_final[, y_cols, drop = FALSE]) # cols align with 'targets'
  )
}
```

```
# final hyperparameters (fixed, no CV here)
p_rf_short <- 12 # from RF CV for h = 1-3
p_rf_long <- 5 # from RF CV for h = 1-12

num_trees_final <- 500 # stable RF size (same scale as RF_Final)
```

```
rf_direct_h <- function(df, predictors, target, p, h,
                        num.trees = num_trees_final) {

  panel <- build_lagged_panel(df, predictors, target, p, h)
  X <- panel$X
  y <- panel$Y[, 1] # single target

  # mtry = sqrt(number of predictors)
  mtry_val <- max(1L, floor(ncol(X) / 3))

  train_df <- data.frame(y = y, X)
```

```

rf_fit <- ranger::ranger(
  dependent.variable.name = "y",
  data                    = train_df,
  num.trees               = num.trees,
  mtry                    = mtry_val,
  write.forest            = TRUE,
  respect.unordered.factors = "order"
)

X_last <- tail(X, 1)
pred   <- predict(rf_fit, data = as.data.frame(X_last))$predictions

as.numeric(pred)
}

targets_to_run <- c("GDP_per_w", "CPI_ALL_CAN", "NHOUSE_P_CAN")

all_short_rf <- list()
all_long_rf  <- list()

# loop over each target
for (tt in targets_to_run) {
  # short: h = 1, 2, 3
  fc_s <- sapply(1:3, function(h) {
    rf_direct_h(df_ml, features_large, tt, p_rf_short, h)
  })
  all_short_rf[[tt]] <- fc_s

  # long: h = 1,...,12
  fc_l <- sapply(1:12, function(h) {
    rf_direct_h(df_ml, features_large, tt, p_rf_long, h)
  })
  all_long_rf[[tt]] <- fc_l
}

last_date <- max(df_ml$Date)

rf_short_all <- tibble(
  variable = rep(targets_to_run, each = 3),
  h         = rep(1:3, times = length(targets_to_run)),
  date      = rep(seq(last_date %m+% months(1),
                      by   = "1 month",
                      length.out = 3),
                  times = length(targets_to_run)),
  forecast = unlist(all_short_rf)
)

rf_long_all <- tibble(
  variable = rep(targets_to_run, each = 12),
  h         = rep(1:12, times = length(targets_to_run)),
  date      = rep(seq(last_date %m+% months(1),
                      by   = "1 month",
                      length.out = 12),

```

```

        times = length(targets_to_run)),
    forecast = unlist(all_long_rf)
)

knitr::kable(
  rf_short_all,
  digits = 4,
  caption = "Direct RF forecasts (ALL VARIABLES, h = 1-3)"
)

```

Table 1: Direct RF forecasts (ALL VARIABLES, h = 1–3)

variable	h	date	forecast
GDP_per_w	1	2025-08-01	0.0028
GDP_per_w	2	2025-09-01	0.0027
GDP_per_w	3	2025-10-01	0.0029
CPI_ALL_CAN	1	2025-08-01	0.0025
CPI_ALL_CAN	2	2025-09-01	0.0027
CPI_ALL_CAN	3	2025-10-01	0.0027
NHOUSE_P_CAN	1	2025-08-01	-0.0007
NHOUSE_P_CAN	2	2025-09-01	-0.0006
NHOUSE_P_CAN	3	2025-10-01	-0.0004

```

knitr::kable(
  rf_long_all,
  digits = 4,
  caption = "Direct RF forecasts (ALL VARIABLES, h = 1-12)"
)

```

Table 2: Direct RF forecasts (ALL VARIABLES, h = 1–12)

variable	h	date	forecast
GDP_per_w	1	2025-08-01	0.0025
GDP_per_w	2	2025-09-01	0.0023
GDP_per_w	3	2025-10-01	0.0027
GDP_per_w	4	2025-11-01	0.0029
GDP_per_w	5	2025-12-01	0.0030
GDP_per_w	6	2026-01-01	0.0027
GDP_per_w	7	2026-02-01	0.0026
GDP_per_w	8	2026-03-01	0.0026
GDP_per_w	9	2026-04-01	0.0028
GDP_per_w	10	2026-05-01	0.0027
GDP_per_w	11	2026-06-01	0.0028
GDP_per_w	12	2026-07-01	0.0028
CPI_ALL_CAN	1	2025-08-01	0.0023
CPI_ALL_CAN	2	2025-09-01	0.0026
CPI_ALL_CAN	3	2025-10-01	0.0024
CPI_ALL_CAN	4	2025-11-01	0.0024
CPI_ALL_CAN	5	2025-12-01	0.0023
CPI_ALL_CAN	6	2026-01-01	0.0023

variable	h	date	forecast
CPI_ALL_CAN	7	2026-02-01	0.0024
CPI_ALL_CAN	8	2026-03-01	0.0025
CPI_ALL_CAN	9	2026-04-01	0.0025
CPI_ALL_CAN	10	2026-05-01	0.0028
CPI_ALL_CAN	11	2026-06-01	0.0031
CPI_ALL_CAN	12	2026-07-01	0.0025
NHOUSE_P_CAN	1	2025-08-01	-0.0008
NHOUSE_P_CAN	2	2025-09-01	-0.0008
NHOUSE_P_CAN	3	2025-10-01	-0.0005
NHOUSE_P_CAN	4	2025-11-01	-0.0006
NHOUSE_P_CAN	5	2025-12-01	-0.0004
NHOUSE_P_CAN	6	2026-01-01	-0.0003
NHOUSE_P_CAN	7	2026-02-01	-0.0003
NHOUSE_P_CAN	8	2026-03-01	-0.0002
NHOUSE_P_CAN	9	2026-04-01	-0.0002
NHOUSE_P_CAN	10	2026-05-01	0.0000
NHOUSE_P_CAN	11	2026-06-01	0.0001
NHOUSE_P_CAN	12	2026-07-01	0.0000

```
# export CSVs for later comparison
write.csv(rf_short_all, "RF_all_variables_short.csv", row.names = FALSE)
write.csv(rf_long_all, "RF_all_variables_long.csv", row.names = FALSE)
```